STRABISMUS

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DEFINITION

- Misalignment of the visual axis hence eyes look in different direction
- Crossed eyes
- Visual axis [line of vision] passes from fovea to object of regard

LEFT EYE SQUINT

BINOCULAR REFLEXES

- Complex set of Reflexes that keep the 2 eyes straight
- Involve both sensory and motor parts of the nervous system
- Most squints occur when something goes wrong in these reflexes

CONT.

- Sensory reflex: takes the 2 images produced by the 2 eyes join (fuses) them together at the brain - 1 stereoscopic image
- Motor reflex: makes sure the 2 eyes always move together (always pointing in the same direction)

NB; reflexes weaker in children – squints more in early childhood

Causes of squints

- 1. sensory factors - disorders of vision
- 2. motor factors - disorders of eye movement
- 3.central factors

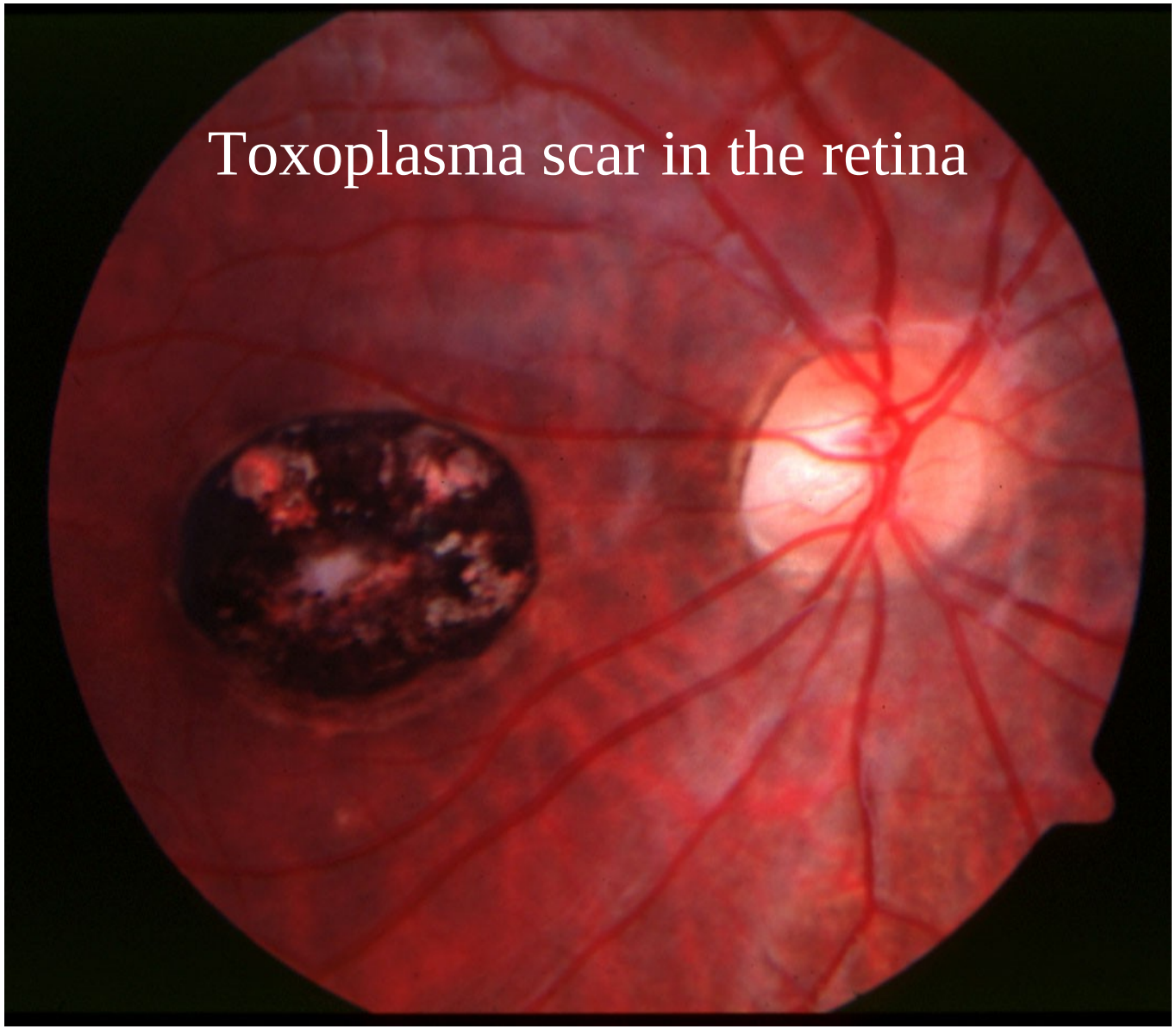
1. sensory factors- Disorders of vision

- The diseased eye forms a blurred image, brain unable to fuse it with the image from the normal eye – hence no stimulus for the diseased eye to remain straight

Diseases –

DISEASES

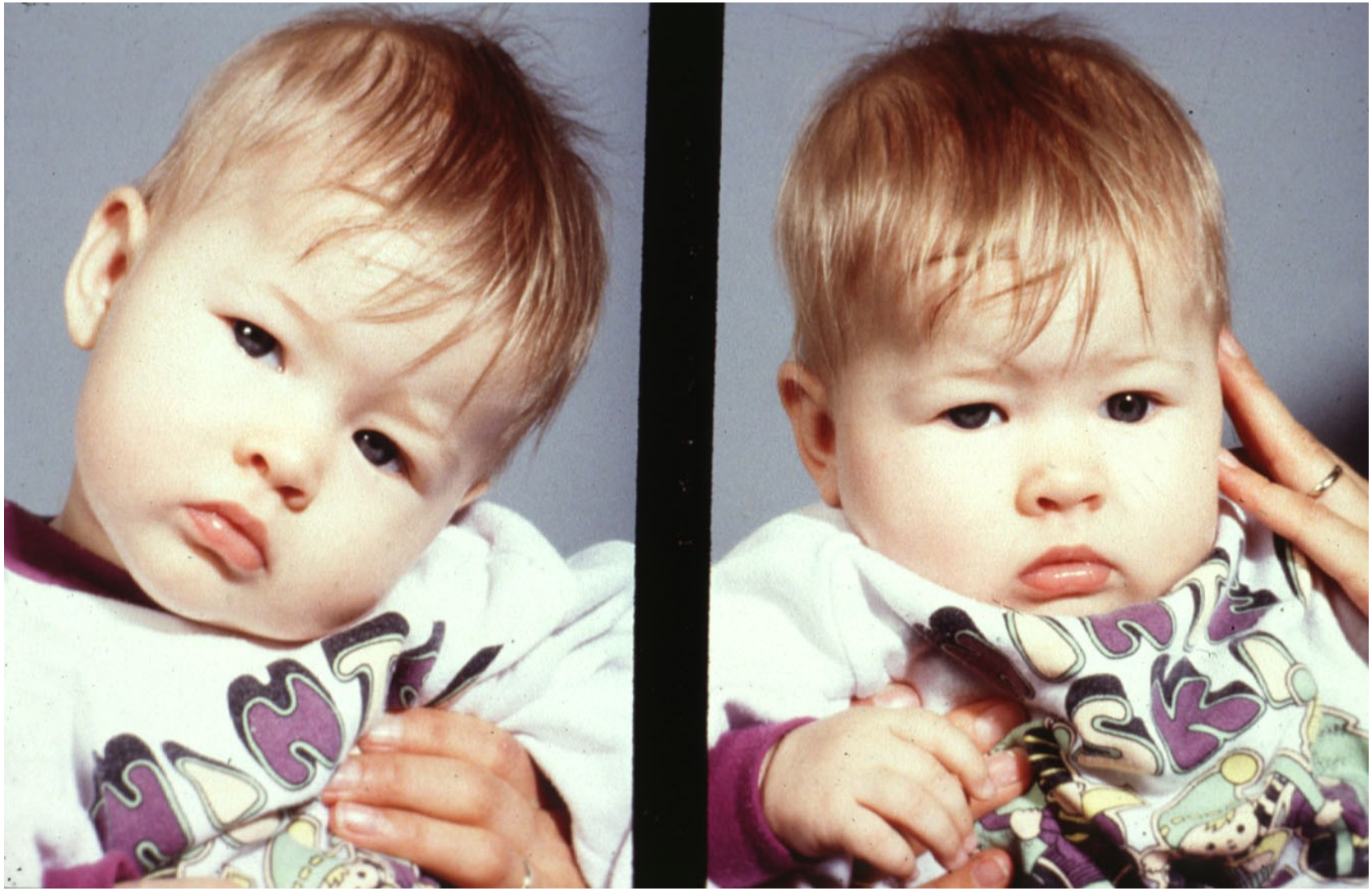
- Media opacities e.g. cataract, cornea opacities
- Refractive errors
- Retinal neuro-disturbances
- Monocular period


Toxoplasma scar in the retina

2. MOTOR FACTORS

- Defective eye movements → defect can be in EOM, nerve
- Developmental anomalies → orbit, muscles, nerves
- Trauma → birth, accidents
- Diseases → myasthenia gravis, Graves disease, Multiple sclerosis, Vascular disease
- Infections → Meningitis, encephalitis, mastoiditis
- Neoplasm → increased ICP

Hyper- tropia (congenital superior oblique muscle palsy)



Left eye 3rd cranial nerve palsy

3. Central factors

- Squint occurs without sensory, motor defect or refractive error
- Occurs during Physical or mental illness, tiredness, emotional stress – results in loss of control of fusion

Effects of squint in early childhood

Short-lived double vision- brain ignores the image from the squinting eye and diplopia stops

- Reduction in vision in that eye- amblyopia
- Suppression
- Degeneration of fusion power

Effects of squint in adults

Squint developing in adulthood results in

- Double vision
- confusion
- abnormal head posture

Prefers to cover or close one eye or turn the head, face

Classification of squints

Depends on

- where the eye faces – convergent, divergent, vertical
- Cause – paralytic, non-paralytic
- Constant or Intermittent

Convergent squint

Convergent squint

Divergent squint

Clinical examination & assessment for squint

- History
- Visual assessment r/o reduced vision
- Measurement of deviation –Cornea light Reflex (Hirschberg test]
- Cover tests, prisms

Central, symmetrical corneal reflexes

Clinical evaluation

- EOM movements
- Refraction
- Full eye examination and Fundoscopy
- Binocular tests

BOYS

Principles of Management of strabismus in children

- Squints of sudden onset, paralytic r/o neurological & medical diseases first
- 1. Refraction and correction of refractive error

Principles of Management of strabismus

2. Improve reduced vision in children thro Amblyopia therapy

Amblyopia therapy

Management of strabismus in children

- 3. Surgical treatment

Aim is to improve cosmetic appearance

Tightening and weakening EOM

NB; usually patient will not recover binocular vision after surgery

Management of squints in adults

- 1. If squint started in childhood – do cosmetic squint surgery

Squint of recent onset- r/o neurological & medical diseases

- 2. Diplopia – cover one eye
- 3. Small squints- correctable with prisms
- 4. Large squints – squint surgery

summary

- A child with a squint needs thorough eye examination
- A squint in a child can cause amblyopia which is treatable if detected early
- Squint of sudden onset may be a sign of neurological/ medical disease
- Management of squint in children is different from adults

references

- 1. Diseases of hot climate by sandford smith
- 2. handbook of orthoptic principles by T. keith lyle
- 3. paediatric opththalmology & strabismus by keneth Wright

Thank you